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COULD WALMART BE A SECRET SOCIALIST PLOT?

Could Walmart be a secret socialist plot? This is, in effect, the question that Fredric Jameson, American literary critic, Marxist political theorist, and cheeky devil, all too briefly poses in a footnote to his 2005 volume *Archaeologies of the Future*, a discussion of the nature of utopia in the age of globalization. Since the demise of post-war technological optimism in the ’70s, Jameson finds that the once robust tradition of utopian thinking has waned considerably; the bare handful of fresh utopias that he identifies as having emerged—be it cyberpunk aesthetics or the cheerleading for corporate-led globalization—appear wholly unimaginative in comparison to their promethean and modernist predecessors, who did not content themselves to merely transform realms of communication and information the way contemporary utopians might. These are mere sectoral concerns, he says, rather than grander, society-wide ambitions; they are not comprehensive utopias.

Today’s soi-disant utopians take little advantage of the genuinely new, what Jameson calls “properly utopian,” resources available now. Bits of a better world that we could exploit are already sprouting up yet no one seems to have noticed them. In a brief footnote, gleefully poking at the progressive consensus that regards Walmart as a globe-barnacling chain of retail hypermarkets, the Galactus of capitalism, the beau idéal—perhaps more so

even than Goldman Sachs—of everything that is wrong with everything that is wrong, Jameson wonders whether we might in fact be missing a trick about this transcontinental marvel of planning and logistics:

> The literary utopists have scarcely kept pace with the businessmen in the process of imagination and construction … ignoring a global infrastructural deployment in which, from this quite different perspective, the Walmart celebrated by Friedman becomes the very anticipatory prototype of some new form of socialism for which the reproach of centralization now proves historically misplaced and irrelevant. It is in any case certainly a revolutionary reorganization of capitalist production, and some acknowledgment such as "Waltonism" or "Walmartification" would be a more appropriate name for this new stage.

But beyond these comments, the provocation is not fully developed. He lets the suggestion just hang there until the publication five years later of an essay on the subject: "Walmart as Utopia." Here, he insists more full-throatedly that Walmart is not merely a useful institution from which, "after the revolution," progressives could (per Lenin) "lop off what capitalistically mutilates this excellent apparatus." It is not residual of the old society, he says, but rather something truly emergent of the new one yet to be born. Walmart is "the shape of a Utopian future looming through the mist, which we must seize as an opportunity to exercise the Utopian imagination more fully, rather than an occasion for moralizing judgments or regressive nostalgia."

This is no contrarian "edgelord" rustling of jimmies, performed simply for the lulz; Jameson is genuinely fascinated with the emergence of this novel entity that is resistant to easy categorization. He compares it to the discovery of a new species of organism, or of a new strain of virus. He delights at the apparent contradiction of how the largest company in the world, even in its full-spectrum dominance—indeed precisely because of this omnipotence—is described by admiring, horrified business writers as a boa constrictor slowly but inexorably strangling market capitalism.

But even here, Jameson is still mostly interested in using Walmart as a thought experiment—a demonstration of "the dialectical character of the new reality," and an example of the notion within dialectics of the unity of

opposites: the firm as “the purest expression of that dynamic of capitalism which devours itself, which abolishes the market by means of the market itself.”

Such philosophical arabesques are more than worthwhile, but we are curious about something perhaps a measure more concrete. We want to take Jameson’s provocation beyond a footnote or a thought experiment and, in the light of what we know about Walmart’s operations, revisit a nearly century-old argument between those who favored socialism and those who asserted that capitalism offered the best of all possible worlds. For beneath the threadbare cliché of the maxim that socialism is “fine in theory, but impossible in practice,” there in fact lie claims about economic planning, and about how to calculate an egalitarian distribution of goods and services without need for markets. Furthermore, the appearance that these claims have been settled by the failure of the undemocratic Soviet Union and its satellites is merely superficial. And counterintuitive as it may seem at first, the no less undemocratic Walmart, and a handful of other examples we will consider, offer powerful encouragement to the socialist hypothesis that a planned economy—this time democratically coordinated by ordinary working people, rather than by bureaucrats or bosses—is not merely feasible, but more efficient than the market.

But before we begin to explain how Walmart is the answer, we first have to ask: What is the question?

The Socialist Calculation Debate

Since the neoliberal revolution of the 1970s and its acceleration following the end of the Cold War, economic planning at scale has been widely derided from right to center-left, and planned endeavors such as public healthcare have been under attack from marketization in most countries. In most jurisdictions, the electricity systems that were once in public hands have long since been privatized; therefore governments committed to efforts to decarbonize electricity companies have had little choice but to employ market mechanisms such as emissions trading or carbon taxation, rather than reducing greenhouse gas emissions via democratic fiat—that is, simply ordering the electricity provider to switch to non-emitting fuel sources. Almost everywhere, transportation, communication, education, prisons,

policing and even emergency services are being spun off wholly or in part from the public sector and provided instead by market actors. Only the armed forces remain a state monopoly, and here only up to a point, given the rise of private security multinationals such as the notorious G4S and Blackwater (rebranded as Academi since 2011). The handful of social democratic and liberal parties that still defend public healthcare and public education do so while making vague assertions that "government has a role to play" or that "government can be a force for good." But they don't really say *why*; and in any case, this is making a case more for the state, rather than for planning per se, even though "the state" and "planning" are far from synonymous. Social democrats today will argue for a mixed economy, or for a mixture of state planning and the market—but again, they do so without saying why. If planning is superior, then why not plan everything? But if some goods and services are better produced by the market than by planning, then what are the attributes of these particular goods and services that make them so? All this activity and argument empty of actual argument reflects a set of policies enacting surrender to an unchangeable status quo, the architects of which only retroactively attempt to transform such capitulation into a coherent ideology. For much of social democracy in the twenty-first century, beliefs follow from policies, rather than policies from beliefs. And while those centrists and conservatives who cheerlead the market stop short of advocating a world where *everything* is allocated via markets, they still do not offer arguments explaining why their preferred admixture of market and planning is superior. When challenged, they simply describe the current state of affairs: "No economy is completely planned or completely market-based." Well, plainly this is true. But again, this gives no explanation as to why their favored configuration is optimal.

Perhaps this is understandable. The endeavors that epitomized planning—those of the Soviet Union and its satellites—collapsed in the face of popular opposition, economic stagnation, a militarily superior geopolitical rival, and a leadership that had all but ceased to believe in its own system. The other major Stalinist power, the People's Republic of China, steered away from state ownership, liberalized its economy, and is now the world's second superpower, while what remains of other Communist-with-a-capital-C states such as Vietnam and Cuba are following in China's wake. It seems, at first glance, almost manifest that the market won the Cold War and that planning lost.

Yet if the market is conclusively, unassailably, incontestably the optimum mechanism for the allocation of goods and services, then why have the economies of Western nations continued to experience mismatches between what is produced and what is required—mismatches that have led to severe recessions and near-catastrophic economic crises since 1991? Why was the global economy barely (and likely temporarily) saved from a Depression-scale collapse in investment in 2008, not by market mechanisms, but as a result of (modest) Keynesian pump priming? What is the source of economic stagnation since the Great Recession? Why, after three decades of steady decreases in inequality in the West in the post-war period leading up to the 1970s, has inequality in the developed countries grown over the last forty years, triggering an explosion in popular anger, along with hard-right reaction, in country after country? Why has infrastructure crumbled and innovation stalled? Why can't the market resolve what is perhaps the greatest threat to modernity, microbial resistance to antibiotics—a situation that risks casting medicine back to the Victorian era—whereas a public sector effort likely could? And why can't the market, left to its own devices, meet the civilizationally existential challenge of climate change?

So the question of market versus planning should appear as unresolved as ever.

In the early decades of the last century, the question of whether the market or planning is the optimal mechanism for the allocation of goods and services was widely accepted as unanswered. In the 1920s and 1930s, left-wing economists influenced by Marxism, on the one hand, and right-wing economists of the neoclassical Austrian School, on the other, were engaged in a vigorous discussion—subsequently known as the "economic calculation problem," or the "socialist calculation debate"—over whether economic planning at scale was feasible. At the time, neoclassicals were not arguing from a position of ideological hegemony. The Soviet Union had recently been established, and the war efforts of both the Allies and the Central Powers were expansive exercises in central planning. By the 1930s, the Bolsheviks had rapidly launched a feudal Russia into electrified, industrial modernity, while few outside the country were aware of the extent of Stalin's crimes, meaning economists who would criticize planning would have to counter what appeared to be substantial evidence in its favor. As a result, partisans on both sides took the idea of planning seriously, and the Austrians had to work hard to try to prove their point, to show how economic planning

was an impossibility.

Viennese mathematician, positivist philosopher and political economist Otto Neurath instigated the “calculation debate” in a series of articles following his experiences as head of the Department of War Economy in the German Empire’s War Ministry. A polymath who had studied mathematics, physics, philosophy and history, his doctorate was in the history of economics, alighting in particular on the non-monetary economy of ancient Egypt. An investigation of the 1912–13 Balkan Wars led him to conclude that war economies are “economies in kind,” or what he termed “natural economies.” Natural economies are those in which money and markets play no role in the allocation of goods; there is no common unit of calculation, no price, and accounting instead takes place in terms of a good or service’s usefulness, described vis-àvis the magnitude of its physical properties. Neurath was also impressed by the ministry’s extensive use of planning during the Great War.

During the socialist November Revolution of 1918, which would succeed at toppling the German Empire, Neurath helped draft a plan for the socialization of the economy of Saxony. Although never a prominent theoretician with the Austrian Social Democratic Party, he believed the upheaval would give him a chance to attempt a practical application of his ideas. He gave speeches on his concepts to mass meetings of miners in the south of the German province, speeches described by his friend and collaborator Wolfgang Schumann as “triumphal processions.” While at this time many left-wing political groups vied for power across Europe, few of them, including the Bolsheviks in Russia, had developed any schemes for the construction of a socialist society beyond slogans calling for an overthrow of the bosses and the free association of the producers. Now that the bosses were gone, how would this free association work? Many called for socialism, but few could describe in detail what that might look like. Neurath, however, began to venture beyond slogans and yearnings to give concrete form to socialism. As a result of the impression Neurath had made, the Social Democratic president of Bavaria, Johannes Hoffmann, asked him to craft and implement a central planning office for this region as well. Upon arrival, he found little more than chaos: no staff, no office, not even a typewriter. Nevertheless, Neurath and his collaborators managed to produce the first working units for economic planning, along with more leaflets and lectures to popularize the concepts. Soon after, revolutionaries declared Bavaria a soviet

(or “council”) republic, but the experiment was short lived. In May 1919, the rightist mercenaries of the Freikorps—a precursor of the Nazis—entered Munich and crushed the council government, killing some 1,000 in ferocious street fighting and a further 700 via summary execution. Neurath was arrested and condemned to eighteen months imprisonment in Germany, but he was ultimately spared in an exchange with the Austrian government orchestrated by Austria’s then–Social Democratic foreign secretary (and Marxist theoretician), Otto Bauer.

Neurath continued to be an active participant in Viennese socialist politics long after the defeat of the Bavarian soviet, participating in the development of adult education and the city’s famously successful experiments in social housing. But ultimately he would become best known for cofounding the Vienna Circle, a group of like-minded scientists and philosophers who contributed to the philosophical movement of logical positivism—in essence an update of nineteenth-century positivism (the assertion that all authoritative knowledge is the product of sensory experience interpreted through reason) that argued for a “scientific conception of the world.” He also became known for a concept he called the “unity of science”: the idea that common scientific laws apply everywhere and at all levels of organization, including the social and even artistic ones.

But this call for a conciliation between different fields of knowledge was no defeatist departure from the political realm, still less from his notions of socialist economic planning. Neurath’s plans for full socialization had been built upon theories of natural (non-monetary) economies and sought to bring different types of knowledge together in order to understand and predict the complexities of the social realm—“empirical sociology,” as he described it. In order to achieve economic efficiency while avoiding social inequity, the organizing structure of the new society would have to be rigorously scientific in its predictions of socioeconomic interactions. In other words, Neurath’s argument for the “unity of science” flowed out of his recognition of the informational needs of nonmarket economies.

But while Neurath’s economic ideas today rest little better than obscure, Ludwig von Mises, Austrian School economist and hero of latter-day neoliberals, took them as deadly serious, in so doing, launching the first counter-volley of the calculation debate. In the seminal 1920 essay “Economic Calculation in the Socialist Commonwealth,” Mises went beyond what by this period was already a longstanding ethical argument against

socialism: that under such a system, there would be no incentive to work and therefore no drive toward excellence. In this short text, Mises instead posed the following questions: In any economy larger than the primitive family level, how could socialist planning boards know which products to produce, how much of each should be produced at each stage, and which raw materials should be used and how much of them? Where should production be located, and which production process was most efficient? How would they gather and calculate this vast array of information, and how could it then be retransmitted back to all actors in the economy? The answer, he said, is that the mammoth scale of information needed—for producers, consumers and every actor in between, and for every stage and location of production of the multitude of products needed in society—is beyond the capacity of such planning boards. No human process could possibly gather all the necessary data, assess it in real time, and produce plans that accurately describe supply and demand across all sectors. Therefore, any economy the size of an entire country that tried to replace the myriad decisions from the multitude of sovereign consumers with the plans of bureaucrats working from incorrigibly flawed data would regularly produce vast, chasm-like mismatches between what is demanded and what is supplied. These inefficiencies would result in such social and economic barbarities—shortages, starvation, frustration and chaos—that even if one accepts the inevitability of inequalities and attendant myriad other horrors of capitalism, the market will still appear benign by comparison.

Meanwhile, Mises argued that the extraordinarily simple mechanism of prices in the market, reflecting the supply and demand of resources, already contains all this information. Every aspect of production—from the cost of all inputs at all times, to the locations of inputs and products, and the changing demands and taste of purchasers—is implicitly captured by price.

But if prices in the market are so much more uncomplicated, effortless and manageable, then why don’t we just stick with them?

Mises’s argument in his 1920 essay, later developed through a series of books, is described to this day by his acolytes as his masterpiece. And not without reason: it is perhaps the strongest argument ever mounted against the idea of socialism. How, indeed, could we replace prices with planning boards? And isn’t socialism supposed to be direct rule by the workers, rather than a replacement of unelected bosses with remote bureaucrats? If centralized by bureaucrats, how could all that information be gathered? And

if decentralized, how could all those millions (and globally, billions) of workers democratically coordinate production decisions?

Neurath, for his part, insisted that prices in the market, as descriptors for behavior in an economy, are no less corrupted by this loss of fidelity because they fail to capture sufficient information on the material circumstances of citizens and fail to describe adequately all the costs or benefits of actions. In a system with market-based provision of healthcare, for instance, price does not describe information on inability to access healthcare, just as price does not reflect the impact of greenhouse gas emissions on the average temperature of the planet.

There is much more to the calculation debate, and we'll briefly outline some of the additional mathematical and computational aspects later on, but for now this theoretical standoff should suffice. It is enough to know that as a result of this impasse, depending on our political persuasions, we have opted either for the information imperfections of the market, or for the information imperfections of planning, without ever resolving the debate. The stalemate could even be tweeted in less than 140 characters: "What about data imperfections leading to shortages?" "Oh yeah? Well what about data imperfections leading to injustices?"

Thus we are stuck. Or so it has seemed for a long time.

Planning in Practice

Mises appeared to many to have turned on its head the aphorism that "Socialism works in theory, but not in practice." He convinced many that planning did not even work in theory. The calculation problem appeared to be socialism's theoretical Achilles' heel.

If something works in theory but not in practice, then there is usually something wrong with the theory. But it is equally true that if something in theory does not work, but in practice it does, then again, something must be wrong with the theory. And here is where the villainous Walmart enters our story. Walmart is perhaps the best evidence we have that while planning appears not to work in Mises's theory, it certainly does in practice. And then some.

Founder Sam Walton opened his first store, Wal-Mart Discount City, on July 2, 1962, in the non-city of Rogers, Arkansas, population 5,700. From

that clichédly humble, East Bumphuck beginning, Walmart has gone on to become the largest company in the world, enjoying eye-watering, People's Republic of China–sized cumulative average growth rates of 8 percent during its five and a half decades. Today, it employs more workers than any other private firm; if we include state enterprises in our ranking, it is the world's third-largest employer after the US Department of Defense and the People's Liberation Army. If it were a country—let's call it the People's Republic of Walmart—its economy would be roughly the size of a Sweden or a Switzerland. Using the 2015 World Bank country-by-country comparison of purchasing-power parity GDP, we could place it as the 38th largest economy in the world.

Yet while the company operates within the market, internally, as in any other firm, everything is planned. There is no internal market. The different departments, stores, trucks and suppliers do not compete against each other in a market; everything is coordinated. Walmart is not merely a planned economy, but a planned economy on the scale of the USSR smack in the middle of the Cold War. (In 1970, Soviet GDP clocked in at about $800 billion in today's money, then the second-largest economy in the world; Walmart's 2017 revenue was $485 billion.)

As we will see, Walmart's suppliers cannot really be considered external entities, so the full extent of its planned economy is larger still. According to *Supply Chain Digest*, that business-management periodical more engrossing than a *Vice* exposé on the furry-fetish web-porn habits of the leadership of ISIS, Walmart stocks products from more than seventy nations, operating some 11,000 stores in twenty-seven countries. TradeGecko, an inventory-management software firm, describes the Walmart system as "one of history's greatest logistical and operational triumphs." They're not wrong. As a planned economy, it's beating the Soviet Union at its height before stagnation set in.

Yet if Mises and friends were right, then Walmart should not exist. The firm should long since have hit their wall of too many calculations to make. Moreover, Walmart is not unique; there are hundreds of multinational companies whose size is on the same order of magnitude as Sam Walton's behemoth, and they too are all, at least internally, planned economies.

Business writers in awe of the company say that the logistical success is ultimately a product of the obsession of Sam Walton (reputedly an inveterate cheapskate) with cost savings, even minor ones, and his use of this advantage

to lower prices, increase volume, and thus enable still further cost savings via expanding economies of scale. While such cost savings are a necessity for all companies, perhaps Walton's single-minded-ness in this regard played some role beyond the usual. What we can say is that the company made a turn toward modern logistics long before many other large firms, and that it has been a trailblazer in logistics innovations that drive down costs.

In 1970, the company opened its first distribution center, and five years later, the company leased an IBM 370/135 computer system to coordinate stock control, making it one of the first retailers to electronically link up store and warehouse inventories. It may seem strange now, but prior to this time, stores were largely stocked directly by vendors and wholesalers, rather than using distributors. Large retailers sell thousands of products from thousands of vendors. But direct stockage—sending each product directly to each store—was profoundly inefficient, leading to regular over-or understocks. Even smaller retailers, who cannot afford their own distribution centers, today find it more efficient to outsource distribution center functions to a logistics firm that provides this service for multiple companies.

Think of when you go to your favorite indie vinyl-revival record shop, and the eyebrow-raising Jack Black–in–*High Fidelity* shop assistant says they cannot get a particular record because their distributor does not carry that album, and you think to yourself, "But I know this record is available; it came out last month on Hello Kitty Pencil Case Records!"—that's why logistical outsourcing happens. It would be far too expensive in terms of labor costs for one tiny store to be able to maintain a commercial relationship with thousands of record labels, and vice versa; but that store can have a relationship with, say, five distributors, each of whom have a relationship with, say, a hundred labels. The use of distributors also minimizes inventory costs while maximizing the variety that a store can offer, at the same time offering everyone along the supply chain a more accurate knowledge of demand. So while your local shop may not carry albums from Hello Kitty Pencil Case Records, via the banal magic of distributors, your tiny local shop will have a relationship with more record labels than they otherwise could.

In 1988, Procter & Gamble, the detergents and toiletries giant, introduced the stocking technique of continuous replenishment, partnering first with Schnuck Markets, a chain of St. Louis grocery stores. Their next step was to find a large firm to adopt the idea, and they initially shopped it to Kmart, which was not convinced. Walmart, however, embraced the concept, and thus

it was that the company's path to global domination truly began.

"Continuous replenishment" is a bit of a misnomer, as the system actually provides merely very frequent restocking (from the supplier to the distributor and thence the retailer), in which the decision on the amount and the timing of replenishment lies with the supplier, not the retailer. Again, you might be asking, how is this an innovation, and why would it make such a difference? (You might also be asking: Why does it now seem like I'm reading a god-awful, capitalism-fellating airport business book? Suck it up. Socialism is all about logistics, comrade.)

The technique, a type of vendor-managed inventory, works to minimize what businesses call the "bullwhip effect," the free market's kissing cousin to Stalinism's shortage problem. First identified in 1961, the bullwhip effect describes the phenomenon of increasingly wild swings in mismatched inventories against product demand the further one moves along the supply chain toward the producer, ultimately extending to the company's extraction of raw materials. Therein, any slight change in customer demand reveals a discord between what the store has and what the customers want, meaning there is either too much stock or too little.

To illustrate the bullwhip effect, let's consider the "too-little" case (although the phenomenon works identically in either scenario). The store readjusts its orders from the distributor to meet the increase in customer demand. But by this time, the distributor has already bought a certain amount of supply from the wholesaler, and so it has to readjust its own orders from the wholesaler—and so on, through to the manufacturer and the producer of the raw materials. Because customer demand is often fickle and its prediction involves some inaccuracy, businesses will carry an inventory buffer called "safety stock." Moving up the chain, each node will observe greater fluctuations, and thus greater requirements for safety stock. One analysis performed in the 1990s assessed the scale of the problem to be considerable: a fluctuation at the customer end of just 5 percent (up or down) will be interpreted by other supply chain participants as a shift in demand of up to 40 percent.

Just like the wave that travels along an actual bullwhip following a small flick of the wrist, a small change in behavior at one end results in massive swings at the other. Data in the system loses its fidelity to real-world demand, and the further you move away from the consumer, the more unpredictable demand appears to be. This unpredictability in either direction is a major

contributing factor to economic crisis as companies struggle (or fail) to cope with situations of overproduction, having produced much more than they predicted would be demanded and being unable to sell what they have produced above its cost. Insufficient stock can be just as disruptive as overstock, leading to panic buying, reduced trustworthiness by customers, contractual penalties, increased costs resulting from training and layoffs (due to unnecessary hiring and firing), and ultimately loss of contracts, which can sink a company. While there is of course a great deal more to economic crisis than the bullwhip effect, the inefficiencies and failures produced by the bullwhip effect can be key causes, rippling throughout the system and producing instability in other sectors. Even with modest cases of the bullwhip effect, preventing such distortions can allow reduced inventory, reduced administration costs, and improved customer service and customer loyalty ("The product you want is right here, ma'am! No need to keep checking other stores! You can always trust us to have what you want. Make sure you come back to us first next time!"), ultimately delivering greater profits.

But there's a catch—a big one for those who defend the market as the optimal mechanism for allocation of resources: the bullwhip effect is, in principle, *eliminated* if all orders match demand perfectly for any period. And the greater the transparency of information throughout the supply chain, the closer this result comes to being achieved. Thus, planning, and above all trust, openness and cooperation along the supply chain—rather than competition—are fundamental to continuous replacement. This is not the "kumbaya" analysis of two socialist writers; even the most hard-hearted commerce researchers and company directors argue that a prerequisite of successful supply chain management is that all participants in the chain recognize that they all will gain more by cooperating as a trusting, information-sharing whole than they will as competitors.

The seller, for example, is in effect telling the buyer how much he will buy. The retailer has to trust the supplier with restocking decisions. Manufacturers are responsible for managing inventories in Walmart's warehouses. Walmart and its suppliers have to agree when promotions will happen and by how much, so that increased sales are recognized as an effect of a sale or marketing effort, and not necessarily as a big boost in demand. And all supply chain participants have to implement data-sharing technologies that allow for realtime flow of sales data, distribution center withdrawals and other logistical information so that everyone in the chain can

rapidly make adjustments.

We hear a lot about how Walmart crushes suppliers into delivering at a particular price point, as the company is so vast that it is worth it from the supplier's persepective to have the product stocked by the store. And this is true: Walmart engages in what it calls "strategic sourcing" to identify who can supply the behemoth at the volume and price needed. But once a supplier is in the club, there are significant advantages. (Or perhaps "in the club" is the wrong phrasing; "once a supplier is assimilated by the Walmart-Borg" might be better.) One is that the company sets in place long-term, high-volume strategic partnerships with most suppliers. The resulting data transparency and cross–supply chain planning decrease expenditures on merchandising, inventory, logistics, and transportation for all participants in the supply chain, not just for Walmart. While there are indeed financial transactions within the supply chain, resource allocation among Walmart's vast network of global suppliers, warehouses and retail stores is regularly described by business analysts as more akin to *behaving like a single firm*.

Flipping all this around, Hau Lee, a Stanford engineering and management science professor, describes how the reverse can happen within a single firm, to deleterious effect. Volvo at one point was stuck with a glut of green cars. So the marketing department came up with an advertising and sales wheeze that was successful in provoking more purchases by consumers and reducing the inventory surplus. But they never told manufacturing, and so seeing the boost in sales, manufacturing thought there had been an increase in demand for green cars and cranked up production of the very thing that sales had been trying to offload.

The same phenomenon occurs in retail as much as it does manufacturing (and manufacturing is merely another link within the retail supply chain anyway), with Toyota being one of the first firms to implement intra- and inter-firm information visibility through its Walmart-like "Kanban" system, although the origin of this strategy dates as far back as the 1940s. While Walmart was pivotal in development of supply chain management, there are few large companies that have not copied its practices via some form of cross–supply chain visibility and planning, extending the planning that happens within a firm very widely throughout the capitalist "marketplace."

Nevertheless, Walmart may just be the most dedicated follower of this "firmification" of supply chains. In the 1980s, the company began dealing directly with manufacturers to reduce the number of links within, and to more

efficiently oversee, the supply chain. In 1995, Walmart further ramped up its cooperative supply chain approach under the moniker Collaborative Planning, Forecasting and Replenishment (CPFR), in which all nodes in the chain collaboratively synchronize their forecasts and activities. As technology has advanced, the company has used CPFR to further enhance supply chain cooperation, from being the first to implement company-wide use of universal product bar codes to its more troubled relationship with radio-frequency ID tagging. Its gargantuan, satellite-connected Retail Link database connects demand forecasts with suppliers and distributes real-time sales data from cash registers all along the supply chain. Analysts describe how stockage and manufacture is "pulled," almost moment-to-moment, by the consumer, rather than "pushed" by the company onto shelves. All of this hints at how economic planning on a massive scale is being realized in practice with the assistance of technological advance, even as the wrangling of its infinities of data—according to Mises and his co-thinkers in the calculation debate—are supposed to be impossible to overcome.

Sears's Randian Dystopia

It is no small irony that one of Walmart's main competitors, the venerable, 120-plus-year-old Sears, Roebuck & Company, destroyed itself by embracing the exact opposite of Walmart's galloping socialization of production and distribution: by instituting an internal market.

The Sears Holdings Corporation reported losses of some $2 billion in 2016, and some $10.4 billion in total since 2011, the last year that the business turned a profit. In the spring of 2017, it was in the midst of closing another 150 stores, in addition to the 2,125 already shuttered since 2010—more than half its operation—and had publicly acknowledged "substantial doubt" that it would be able to keep any of its doors open for much longer. The stores that remain open, often behind boarded-up windows, have the doleful air of late-Soviet retail desolation: leaking ceilings, inoperative escalators, acres of empty shelves, and aisles shambolically strewn with abandoned cardboard boxes half-filled with merchandise. A solitary brand-new size-9 black sneaker lies lonesome and boxless on the ground, its partner neither on a shelf nor in a storeroom. Such employees as remain have taken to hanging bedsheets as screens to hide derelict sections from customers.

The company has certainly suffered in the way that many other brick-and-mortar outlets have in the face of the challenge from discounters such as Walmart and from online retailers like Amazon. But the consensus among the business press and dozens of very bitter former executives is that the overriding cause of Sears's malaise is the disastrous decision by the company's chairman and CEO, Edward Lampert, to disaggregate the company's different divisions into competing units: to create an internal market.

From a capitalist perspective, the move appears to make sense. As business leaders never tire of telling us, the free market is the fount of all wealth in modern society. Competition between private companies is the primary driver of innovation, productivity and growth. Greed is good, per Gordon Gekko's oft-quoted imperative from *Wall Street*. So one can be excused for wondering why it is, if the market is indeed as powerfully efficient and productive as they say, that all companies did not long ago adopt the market as an internal model.

Lampert, libertarian and fan of the laissez-faire egotism of Russian American novelist Ayn Rand, had made his way from working in warehouses as a teenager, via a spell with Goldman Sachs, to managing a $15 billion hedge fund by the age of 41. The wunderkind was hailed as the Steve Jobs of the investment world. In 2003, the fund he managed, ESL Investments, took over the bankrupt discount retail chain Kmart (launched the same year as Walmart). A year later, he parlayed this into a $12 billion buyout of a stagnating (but by no means troubled) Sears.

At first, the familiar strategy of merciless, life-destroying post-acquisition cost cutting and layoffs did manage to turn around the fortunes of the merged Kmart-Sears, now operating as Sears Holdings. But Lampert's big wheeze went well beyond the usual corporate raider tales of asset stripping, consolidation and chopping-block use of operations as a vehicle to generate cash for investments elsewhere. Lampert intended to use Sears as a grand free market experiment to show that the invisible hand would outperform the central planning typical of any firm.

He radically restructured operations, splitting the company into thirty, and later forty, different units that were to compete against each other. Instead of cooperating, as in a normal firm, divisions such as apparel, tools, appliances, human resources, IT and branding were now in essence to operate as autonomous businesses, each with their own president, board of directors,

chief marketing officer and statement of profit or loss. An eye-popping 2013 series of interviews by *Bloomberg Businessweek* investigative journalist Mina Kimes with some forty former executives described Lampert's Randian calculus: "If the company's leaders were told to act selfishly, he argued, they would run their divisions in a rational manner, boosting overall performance."

He also believed that the new structure, called Sears Holdings Organization, Actions, and Responsibilities, or SOAR, would improve the quality of internal data, and in so doing that it would give the company an edge akin to statistician Paul Podesta's use of unconventional metrics at the Oakland Athletics baseball team (made famous by the book, and later film starring Brad Pitt, *Moneyball*). Lampert would go on to place Podesta on Sears's board of directors and hire Steven Levitt, coauthor of the pop neoliberal economics bestseller *Freakonomics,* as a consultant. Lampert was a laissez-faire true believer. He never seems to have got the memo that the story about the omnipotence of the free market was only ever supposed to be a tale told to frighten young children, and not to be taken seriously by any corporate executive.

And so if the apparel division wanted to use the services of IT or human resources, they had to sign contracts with them, or alternately to use outside contractors if it would improve the financial performance of the unit—regardless of whether it would improve the performance of the company as a whole. Kimes tells the story of how Sears's widely trusted appliance brand, Kenmore, was divided between the appliance division and the branding division. The former had to pay fees to the latter for any transaction. But selling non-Sears-branded appliances was more profitable to the appliances division, so they began to offer more prominent in-store placement to rivals of Kenmore products, undermining overall profitability. Its in-house tool brand, Craftsman—so ubiquitous an American trademark that it plays a pivotal role in a Neal Stephenson science fiction bestseller, *Seveneves*, 5,000 years in the future—refused to pay extra royalties to the in-house battery brand DieHard, so they went with an external provider, again indifferent to what this meant for the company's bottom line as a whole.

Executives would attach screen protectors to their laptops at meetings to prevent their colleagues from finding out what they were up to. Units would scrap over floor and shelf space for their products. Screaming matches between the chief marketing officers of the different divisions were common

at meetings intended to agree on the content of the crucial weekly circular advertising specials. They would fight over key positioning, aiming to optimize their own unit's profits, even at another unit's expense, sometimes with grimly hilarious result. Kimes describes screwdrivers being advertised next to lingerie, and how the sporting goods division succeeded in getting the Doodle Bug mini-bike for young boys placed on the cover of the Mothers' Day edition of the circular. As for different divisions swallowing lower profits, or losses, on discounted goods in order to attract customers for other items, forget about it. One executive quoted in the Bloomberg investigation described the situation as "dysfunctionality at the highest level."

As profits collapsed, the divisions grew increasingly vicious toward each other, scrapping over what cash reserves remained. Squeezing profits still further was the duplication in labor, particularly with an increasingly top-heavy repetition of executive function by the now-competing units, which no longer had an interest in sharing costs for shared operations. With no company-wide interest in maintaining store infrastructure, something instead viewed as an externally imposed cost by each division, Sears's capital expenditure dwindled to less than 1 percent of revenue, a proportion much lower than that of most other retailers.

Ultimately, the different units decided to simply take care of their own profits, the company as a whole be damned. One former executive, Shaunak Dave, described a culture of "warring tribes," and an elimination of cooperation and collaboration. One business press wag described Lampert's regime as "running Sears like the Coliseum." Kimes, for her part, wrote that if there were any book to which the model conformed, it was less *Atlas Shrugged* than it was *The Hunger Games*.

Thus, many who have abandoned ship describe the harebrained free market shenanigans of the man they call "Crazy Eddie" as a failed experiment for one reason above all else: the model kills cooperation.

"Organizations need a holistic strategy," according to the former head of the DieHard battery unit, Erik Rosenstrauch. Indeed they do. But is not society as a whole an organization? Is this lesson any less true for the global economy than it is for Sears? To take just one example: the continued combustion of coal, oil and gas may be a disaster for our species as a whole, but so long as it remains profitable for some of Eddie's "divisions," those responsible for extracting and processing fossil fuels, these will continue to act in a way that serves their particular interests, the rest of the company—or

in this case the rest of society—be damned.

In the face of all this evidence, Lampert is, however, unrepentant, proclaiming, “Decentralised systems and structures work better than centralised ones because they produce better information over time.” For him, the battles between divisions within Sears can only be a good thing. According to spokesman Steve Braithwaite, “Clashes for resources are a product of competition and advocacy, things that were sorely lacking before and are lacking in socialist economies.”

He and those who are sticking with the plan seem to believe that the conventional model of the firm via planning amounts to communism. They are not entirely wrong.

Interestingly, the creation of SOAR was not the first time the company had played around with an internal market. Under an earlier leadership, the company had for a short time experimented along similar lines in the 1990s, but it quickly abandoned the disastrous approach after it produced only infighting and consumer confusion. There are a handful of other companies that also favor some version of internal market, but in general, according to former vice president of Sears, Gary Schettino, it “isn’t a management strategy that’s employed in a lot of places.” Thus, the most ardent advocates of the free market—the captains of industry—prefer not to employ market-based allocation within their own organizations.

Just why this is so is a paradox that conservative economics has attempted to account for since the 1930s—an explanation that its adherents feel is watertight. But as we shall see in the next chapter, taken to its logical conclusion, their explanation of this phenomenon that lies at the very heart of capitalism once again provides an argument for planning the whole of the economy.

3
ISLANDS OF TYRANNY

Some years before the relatively obscure Marxist critic Frederic Jameson was fashioning utopian visions based on Walmart's internal planning, a much more mainstream figure, economist Herbert Simon, had a not dissimilar realization. A polymath, winner of both the Swedish National Bank's Prize in Economic Sciences in Memory of Alfred Nobel (widely but inaccurately described as the Nobel Prize in Economics) and the Turing Award (widely and comparatively more accurately described as the Nobel Prize for Computing), Simon in 1991 offered up the following thought experiment—one that might have seemed out of place to regular readers of the prestigious but orthodox *Journal of Economic Perspectives*:

> Suppose that [a mythical visitor from Mars] approaches the Earth from space, equipped with a telescope that reveals social structures. The firms reveal themselves, say, as solid green areas with faint interior contours marking out divisions and departments. Market transactions show as red lines connecting firms, forming a network in the spaces between them. Within firms (and perhaps even between them) the approaching visitor also sees pale blue lines, the lines of authority connecting bosses with various levels of workers. As our visitor looked more carefully at the scene beneath, it might see one of the green masses divide, as a firm divested itself of one of its divisions. Or it might see one green object gobble up another. At this

> distance, the departing golden parachutes would probably not be visible.
>
> No matter whether our visitor approached the United States or the Soviet Union, urban China or the European Community, the greater pan of the space below it would be within the green areas, for almost all of the inhabitants would be employees, hence inside the firm boundaries. Organizations would be the dominant feature of the landscape. A message sent back home, describing the scene, would speak of “large green areas interconnected by red lines.” It would not likely speak of “a network of red lines connecting green spots.”

Simon intended his tale of visiting Martians as a light chiding of his fellow economists for ignoring how pervasive authoritarian power relationships and planning actually are under capitalism. Planning was in fact almost everywhere you looked, even though the discipline of economics had largely spun tales even more fantastical than UFOs visiting Earth: the fairy story of a harmonious and self-regulating market economy. Yet there has always been a minority of economists, like Simon, who have dissented, recognizing the pervasiveness, a few even the promise, of planning.

Ronald Coase Asks Around

In the Depression year of 1931, a twenty-year-old British economics student arrived in Chicago to pursue an unusual research project. He was there to study something that at first glance appeared utterly obvious; yet in reality it was anything but. Ronald Coase went to the United States to do something that, up to this point, few scholars in the still-young discipline of economics had cared to do: investigate how the firm, the black box at the heart of the economy, actually operated.

Coase’s question was a simple one, but one to which the economics he had been taught didn’t yet have an answer: “Why are there these ‘islands of conscious power’? … If production is regulated by price movements [and] production could be carried on without any organization at all, well might we ask, why is there any organization?” In other words, if the market is the magic bullet to all human interaction, then even the simplest work tasks—from “stock this shelf” to “format this spreadsheet”—could theoretically be

governed by prices on markets rather than by managers giving orders. Somewhat naively, Coase asked, why isn't everything bought and sold on its own little market? Why are there so many times more Walmarts than there are Sears? Why do companies—from mom-and-pop shops to corporate behemoths—even exist?

Noam Chomsky, the great linguist and lifelong critic of American foreign policy, had a pithy answer: Coase's "islands of conscious power" are also "islands of tyranny." Thus, economists are unwilling to pry open the black box of the firm because it holds capitalism's dirty secret. The market economy is not only rife with planning, but with *authoritarian* planning that concentrates economic decision making in the hands of wealth owners and keeps workers in line. Companies plan everything from how money is distributed between departments to the exact amount of time it should take to assemble a hamburger—and in every case, they plan which individual worker does which task, when, where and how. When you're on the clock, what the boss says goes.

Open nearly any introductory economics textbook, however, and the world appears as a nearly boundless realm of choice. Among the paeans to freedom and to the spontaneous efficiency of markets, few words even graze the everyday planning that goes on within the four walls of the firm. Fewer, still, name it as coercive. Planning under capitalism is about making people do things—without their input and not necessarily in their interest. At best, economists will bring up planning in order to ridicule it, failing, or refusing, to grasp its centrality even in a market system. Coase's seemingly naive questions start us on a path toward a corrective.

Coase, however, was no fellow traveler. While he had even flirted with socialist ideas in his youth, his economics education quickly turned him to the right (sadly, an all too common phenomenon). Coase argued that companies do all of this apparent in-house imitation of the Soviet Union simply because the cost is too high of leaving up to markets every last coordinating decision. This was quite a clever explanation for the dissonance between copious corporate planning within and throughout a free market system. Economists are fond of the saying that "there is no free lunch." Coase applied this to markets themselves. Markets introduce a whole web of what he called "transaction costs." Writing a contract, setting up a market or finding the best price all take up resources and time. So long as the cost of doing all this was cheaper in house than on the market (and it was), it was

only rational to keep it in house. So the “free” market isn’t really free either! Coase argued that it only makes sense that some decisions would be left to planning—a decision is made, and it is done. Planning is more efficient—though for Coase, only up to a certain point. Having completed his tour of American business and witnessed its inner workings, upon his return to Britain, he compiled his thoughts in a 1932 lecture to University of Dundee students little younger than himself, although it would be another five years before he published his results.

The resulting text, “The Nature of the Firm”, features a quote from economist Dennis Robertson—a close collaborator of famed British macroeconomist John Maynard Keynes, and the originator of the concept of the “liquidity trap”—in which Robertson talks of the curiosity of the very existence of companies, unflatteringly describing them as “islands of conscious power in this ocean of unconscious cooperation, like lumps of butter coagulating in a pail of buttermilk.” But where Robertson had merely remarked upon the mystery, Coase explained it: “Those who object to economic planning on the grounds that the problem is solved by price movements can be answered by pointing out that there is planning within our economic system [that] is akin to what is normally called economic planning.”

He was ignored for his insight. To this day, while hats are now tipped to Coase, and even though planning is plainly ubiquitous, taking place at heretofore unimagined scales, most economists talk very little about it. Economics textbooks offer in-depth explanations of consumer goods markets, the labor market, the money market or even the entire economy as one big market, but little to nothing about the planning inside firms. At best, economists will briefly mention planning, and then only to ridicule it. In much of mainstream economics, the firm is just a mathematical equation that consumes inputs and produces outputs. How it does this is rarely asked; its internal workings are insufficiently interesting. Or sufficiently embarrassing.

Willful disregard for the reality of planning is common enough. Adam Smith, the eighteenth-century Scot now considered the father of economics, is famous for introducing the “invisible hand” of the market. By this he meant no mystical force, but the idea that while individuals are making decisions whether to sell or to buy in the pursuit of self-interest, they are “led by an invisible hand to promote an end which was no part of [their] intention”—the welfare of society realized through a market system. Smith’s

hand often appears in economics textbooks as proof that markets produce, without any kind of plan, the best possible outcomes. However, Smith himself understood that real economies involve all manner of nonmarket interactions—even the phrase “invisible hand” makes but a single appearance in his *Wealth of Nations*. Smith, for example, assumed that factory owners would scheme together—that is, plan—to keep wages low. Later economists would concentrate only on the first half of his story: that the market system produces order out of chaos, all on its own.

Yet the vision of an orderly but completely unplanned market economy is nothing but fantasy. Planning exists in the market system and on a truly enormous scale. Today, the volume of transactions carried out within firms is as large as that carried out between them. Managers have always been very concerned with planning, but it is only by diving deeply into practical management texts that we can learn about its extent under capitalism. Economists have hidden it behind a tangled web of seeming disorder.

Even so, the fortress built by mainstream economists over the twentieth century is not so monolithic that it excludes all doubt. The seeds of a critique of the planless world were sown even within its walls. Certainly, Coase was not arguing for planning on a large scale. He was simply a mainstream economist willing to look at the world honestly and notice the core role played by planning and control within capitalist business.

The Calculation Debate Continued

At the same time that Coase was traveling about, asking corporate managers why they didn’t have markets for moving products from shelves at one end of a warehouse to ones at the other, economists elsewhere were still busy arguing whether it was necessary to have markets at all. As noted earlier, Ludwig von Mises argued in 1920 that socialist planning of an entire economy was impossible because complex economies of the kind we now have need both markets and prices. In his view, markets decentralize the vast troves of information that a single planner couldn’t compile and calculate. Prices, however, make it possible to compare vastly different things; without them, he reasoned, how would planners know the relative worth of things as disparate as a car factory and a ballpoint pen and ultimately decide how many of each there should be? The counterargument that best answered these

questions, at least for a while, finally came in 1937 from Polish economist Oskar Lange.

Lange's life and work were full of contradictions. A lifelong socialist and Marxist economist, Lange was equally at home in the minutiae of mainstream neoclassical economics as in the footnotes to Marx's *Capital*. Although he ended up teaching at the Warsaw Higher School of Planning and Statistics during the era of postwar Stalinist dogmatism, Lange also spent time at Harvard in the 1930s and taught in the economics department at the University of Chicago from 1938 to 1945, just as the latter was becoming a bastion of free market orthodoxy. And despite being a proponent of market socialism, Lange nevertheless served the Polish state even in its Stalinist incarnation—first as ambassador to the United States, then as representative to the UN, and finally as member of the council of state. These contradictions, however unlikely, worked in Lange's favor when it came to the calculation debate (see chapter 2).

Lange had read his neoclassical economists. However, he believed their models of the capitalist economy could be commandeered and repurposed for socialist planning. Under capitalism, when H&M makes too many skinny, off-purple corduroy trousers, its stores eventually drive down the price to entice people to buy them. Demand meets supply when the price falls—at least that's what happens in theory. In reality, the extra pants can end up in landfills, and H&M's production for next season can end up moving somewhere with lower wages to make ever lower prices possible. Using the equations of Léon Walras, one of the founders of the neoclassical school, Lange wrote a pamphlet in 1937 that imagined a planned economy, which imitated the market without these downsides. Lange's fictional socialist planners would manipulate "shadow prices" on paper, rather than waiting for real prices to filter down from cash registers to production decisions. Like a UV light at a crime scene, socialist planning would make explicit all the math that only happened in the background in models of capitalism. Lange answered Mises's challenge—that prices and markets were necessary to any economic rationality—by incorporating them into a model of market socialism.

The key was devising how planners would figure out which shadow prices are the right ones—those that ensure the socialist economy is making enough, but not too much, of everything. For this, Lange repurposed another idea from Walras: *tatonnement*. In French, Walras's native tongue, the word

means “groping toward.” Walras imagined that markets groped toward the right prices until they found the holy grail of economics: general equilibrium, where all markets are in balance and the amount supplied of every single good or service is exactly equal to the amount demanded. Add some more math, and mainstream economists will tell you that they’ve proven that everyone is also as happy as can be, living in the best of all possible worlds.

Lange, however, figured planners could actually perform this tatonnement better than markets. Unlike in Otto Neurath’s natural economy (discussed in chapter 2), people under Lange’s market socialism would still go to (government-run) stores to buy consumer goods, signaling to planners what they wanted produced. Producers—all also publicly owned—would aim to produce what the planners translated from consumer demands as efficiently as possible, without needing to leave room for profit after covering costs. As the economy produced things and consumers bought them, central planners would run equations, figure out what there was too much of and what there was too little of, and adjust the “shadow prices” until everything was in sync. Even without all the correct information available at once, Lange’s expected his planners to grope toward equilibrium like markets did under capitalism, only better and faster. And it would only be a matter of time before computers came along that were powerful enough to make the process faster still. Lange spent his final years fascinated by computer science and cybernetics. In one of his last papers, he wrote: “The market process with its cumbersome tatonnements appears old-fashioned. Indeed, it may be considered as a computing device of the pre-electronic age.”

Around the same time that Lange developed his theory of planning, the American economist Abba Lerner was working on his own version of market socialism. The two thinkers complemented each other so well that the idea that socialist planning could replicate capitalist efficiency came to be called the Lange-Lerner theorem. Mimicking parts of the theory of capitalism, Lange and Lerner wanted to show that planning could meet and even exceed capitalism’s own measures for squeezing the most human satisfaction out of scarce resources.

By the time the Second World War began, many classical economists grudgingly admitted that Lange’s arguments worked—at least in theory. If the socialist system of planning Lange and others described was theoretically possible, then the only question that remained was whether it was feasible. Although corporate and military planners, averse to socialism but intrigued

by the power of even the simplest mathematical calculation for resource management and control, were beginning to use crude versions of formalized planning tools, it was difficult to imagine when—if ever—the computing power required for planners to solve Lange's equations in reasonable time on an economy-wide scale would be available. With seemingly dim prospects for viable application, there was no reason to trumpet the fact that the socialists might be right.

Hayek's Riposte

Such defeatism alarmed another Austrian economist, Friedrich von Hayek, who, following in the footsteps of Mises, was determined to prove Lange wrong. Hayek is better known today as the godfather of neoliberalism, the pro-market ideology that has come to dominate government policy around much of the world, the first incarnation of which is best exemplified by the administrations of Margaret Thatcher in the UK and Ronald Reagan in the United States during the 1980s. Hayek was explicit about wanting ideological regime change. The postwar welfare state truce between capital and labor had barely been installed when Hayek joined a small group of right-wing radicals to found the Mont Pelerin Society in 1944—a free market think tank before its time. It was integral to their task of reshaping ideology that they have at the ready a rebuke to Lange, Lerner and the other socialists who looked to have the upper hand.

For someone who believed so fervently in capitalism, Hayek offered a very honest picture of the system. Maybe it was precisely because he was so ideologically committed to capitalism that he could talk about its shortcomings—all the ways it deviated from the fantasies of the neoclassical economists with their perfect humans, perfect markets and perfect information. Hayek questioned these central assumptions. People are not hyperrational—we have incomplete, imperfect ideas about the world. Markets are never quite in sync: there is always too much or too little of something. Capitalism is dynamic, a process of constant change rather than a state of equilibrium. On this last point, Hayek harked back to Marx and Smith. But as we'll see, it would take a few decades for the mainstream of economics to embrace such notions.

We have to admit that Hayek was right in rejecting mainstream fantasies.

In fact, it was Lange who had underestimated the problems he inherited from the economics of his time. Here, he differed from Marx. While Marx undertook a thoroughgoing critique of the classical school, which dominated when he was writing, Lange primarily tried to replace "capitalist" variables in the equations of dominant neoclassical economics with "socialist" ones. In doing so, he took on all the flawed assumptions of the mainstream model. These included everything from an impossibly rational *Homo economicus* to eventual general equilibrium to the "completeness" of markets—meaning a market for every possible thing, at every possible time present and future. (In practice, completeness would mean that you could agree to buy—today, for a firm price—a unit of Amazon stock, a haircut or even a block of aged cheddar to be delivered at any precise point in the future, whether two weeks and three hours from now—or even in fifty years!) These assumptions are not only patently false under even the most extreme variants of capitalism; they would soon be challenged—slowly and cautiously—even by neoclassical economists.

Without this baggage, Hayek took a different tack from the silent but grudging acceptance of the mainstream. He rejected Lange's case outright. Hayek argued that markets—incomplete, permanently off tilt, full of fallible humans—do not just aggregate and calculate information. Markets are producers of information and knowledge. Even if Lange's market socialism allowed planners to calculate better and faster than did free markets, planning would ultimately still be impossible because planners would not have the information created by market interactions to use in their calculations. Buying and selling may not generate technical and scientific knowledge, but it still creates all that knowledge of "time and place" that is instrumental to making efficient production and distribution decisions. Hayek argued that the problem for planners was not in the "how"—the equations to use—but in the "what"—the data that goes into the equations. The copious information planners need is unavailable before markets work their magic. Decentralization creates coordination: only the market can bring together the information that is normally isolated in the heads of different individuals.

Hayek, however, was writing before the advent of "big data," which is testing the limits of just how much granular information can be collected. It seems that he also wrote in blissful ignorance of Coase, who had shown just how flimsy the veneer of decentralized decision making really is, even under capitalist markets.

If Hayek sounds like a radical democrat, the affinity is purely superficial. What he is after is not so much freedom for people, but rather *freedom for information and money*—those two central lubricants of market activity. Human beings, after all, are not capable of democratically coordinating complex systems, so they must therefore submit themselves to the dictates of the market, onboarding its anonymous decisions no matter how profound the social costs it creates. The argument against planning clearly hinges on Hayek's ideological commitments

Oddly enough, despite challenging the market socialists head on, Hayek's ideas were initially ignored, perhaps because they were critical not only of left-wing, but also mainstream economic opinion. At a time when even Richard Nixon was pronouncing that "we are all Keynesians now," how could their maximalist rhetoric be anything but out of step? The debate on the calculation problem continued to unfold in the pages of obscure economic journals. The world, however, had moved on.

But shortly after Nixon's startling declaration of allegiance, the existing economic orthodoxies on both sides of the Berlin Wall were violently thrown into question. By the 1970s, "really existing socialism" was mired in economic crisis, its cracks beginning to show. The "free world" was troubled, too, experiencing its most severe economic crisis of the postwar period. Political and economic elites saw in the crisis an opening to unwind their postwar compromise with labor, a compact borne not of love, but out of their fear of revolution. It was in this context that the new heterodoxy championed by Hayek became efficacious outside the walls of the academy at last.

We've All Been Misinformed

Something incredible happened to the discipline of economics in the 1970s: the professors suddenly discovered human beings were *not* the equivalent of walking calculators. Alongside this revelation, many others among the most cherished beliefs of economics had been cast into potential doubt. Much of the entire mainstream economics project since the late nineteenth century had been built on the foundation of perfectly rational humans. Models of markets working together in seamless harmony, as well as arguments about the market system producing the best outcomes, relied on the pretty fantastical assumption that each of us have any and all information permanently at our

fingertips.

As some economists began to question the notion of hyperrational humans, they found Coase's notion of transaction costs to be a useful concept that could help save the rest of the discipline. The new field of transaction cost economics turned Coase's insights about planning within capitalism into a story about flawed humanity. If our world diverged from one populated by perfectly rational beings, then some nonmarket transactions could be grudgingly admitted into the market system—as long as our imperfections were more costly than the benefits we could get from markets. Even our imperfections could be co-opted into the same story about capitalism as the best of all possible worlds!

However, once the Pandora's box of flawed humanity is opened, it is hard to close. Joseph Stiglitz, another winner of the Swedish National Bank's Prize in Economic Sciences in Memory of Alfred Nobel whom the Left sometimes uses to lend credibility to anti-austerity politics, first made his name by furthering the critique of the assumption of human rationality while still making a case for markets. Distinct from the earlier mythology of a perfectly rational *Homo economicus*—nowhere to be found in reality, but for so long beloved by economists—the economics of information that Stiglitz helped launch started from the seemingly obvious idea that getting our hands on, and using, information is usually costly, and sometimes impossible.

An example economists love to use is the market for private health insurance. There is only so much an insurer can do to see if a person buying insurance is relatively healthy. Developing a better and better picture costs more and more. At some point, the costs prevent further information acquisition from making sense. In the same way, hiring a mechanic to take apart and inspect the engine of a used car to find out if it is a "lemon" can cost more than the car itself. Markets can fail: some people will end up overpaying for health insurance, while others will be uninsured. Your local sketchy used car dealership isn't likely to be the first place you'd think of as a well-functioning market.

Beyond individual markets, Stiglitz and others were asking a bigger question: What if the entire economy was something of a used car dealership? Once enough examples of failing markets accumulate, the entire system's efficiency and justice can be called into doubt. In short, the economics of information ultimately challenges the argument that capitalism, despite its flaws, is the best of all possible worlds. However, rather than

seeing information problems as a reason to explore collective, democratic decision-making alternatives that could bring people and information together, economists went to work making market theory work in spite of humanity's imperfections. Since the '70s, the economics of information has generated ever more ingenious ideas for incentivizing people or organizations to do things—all, of course, within the bounds of capitalist markets.

Mechanism design is one such idea. In this obscure corner of economics, economists drum up—elegant, but often mathematically complex—means to compel people or companies to reveal information that they would otherwise keep secret. A new auction format created by economists in the early 1990s to help the US government sell off cell phone frequencies to telecommunications firms is an exemplary case. The auction had rules designed to force companies to reveal how much the rights to frequencies were really worth to them—lying would see them lose the rights to competitors. The design netted the government hundreds of millions of dollars more than expected and is now commonplace around the world.

Mechanism design is a kind of planning, although a very indirect one. Economic decision making of any kind—whether outright planning or a "designed" market—needs to gather the bits and pieces of information spread between people. But information problems don't preclude other ways of doing things. Rather than creating a complex process that ultimately benefits a few big players, governments today could choose to run a public cell phone utility, which would constitute one more step on the way to greater socialization. As things stand, however, governments make some money on the auction, but give up control over a valuable resource. This also leaves behind a market dominated by a few big players who can charge famously high prices backed by shoddy customer service.

Other mechanisms "align incentives"—for example, trying to ensure that workers internalize and act in accord with the goals of managers or ensure that managers internalize and act in accord with the goals of stock owners. Mechanism design is just one more example showing that the free market also has to be planned. Real-world markets must be consciously made and remade.

Speaking of Making People Do Things …

What makes Coase's theory and the economics of information so important is that they show us why the capitalist planning that takes place all around us is such a blind spot, not just in economics, but in our everyday perception of the world.

Mainstream economics ignores the disciplinary nature of business. It has a lot to say about competition between firms, but it overlooks questions of power within them. The intricate explanations for bringing workers together in firms skirt around a fundamental issue: there is a gulf that separates workers' formal freedom to quit their job, if they don't like it, from the fact that we have to work to survive, and thus do not really have that freedom at all. Workers are brought together under the quite literally despotic rule of managers within businesses (a business is not a democracy) because, fundamentally, we have no other choice. Even in those workplaces where management has offered workers a modicum of control over decision making, outside of strong union representation, this gift of freedom and democracy is offered (and withdrawn) at the pleasure of management. This is the definition of authoritarian, that is, nondemocratic, rule—Chomsky's "islands of tyranny." Too often we confuse the violence of despots with what makes despotism wrong. But much of this violence is a grotesque tool to enforce submission. It is this unfreedom—unchallengeable control of a human by another—that is the worst crime.

Under capitalism, businesses buy the time and the energy of workers, and during that time, they can dispose of workers as they wish (within the bounds of the laws of physics and legal or union constraints imposed as the result of class struggle). One of the few economists before Coase to look inside the black box of business was, as it happens, Karl Marx. Marx saw the firm as an instrument for extracting profit off the backs of workers. He alighted upon a simple fact: workers are paid a wage for their time, not for what they produce. Profit comes from the difference between what a business can pay its workers (plus the cost of materials, themselves made or extracted from the earth by other workers) and the value of what these same workers are able to produce.

Coase thought that firms planned simply to save costs. For Marx, what happens inside firms is much more important: it determines how everything we produce is divided up between us. How we produce goods and services is closely related to how much of what we produce goes to whom. Under capitalism, the class of owners (businesspeople or shareholders) receive

much more relative to the class of producers (workers).

The manager's exercise of central planning over his small province of tyranny is therefore not simply a better means to an end, as Coase thought, but a reflection of how the economy actually works. The adversarial relationship between bosses and workers that capitalism creates is no accident of markets merely introducing transaction costs that are best avoided through planning. Yet for mainstream economists, the confrontation between workers and managers only comes up in the context of "shirking." The GPS device in the UPS driver's truck, the call center badge that monitors washroom breaks or the white-collar worker's app that tracks web browsing history are the sticks requiring one does as one is told; the bonuses are the carrots.

Shirking, however, is a very rational response for someone who has little or no say over their work, often has no deeper sense of collective responsibility and knows that the profit from what they do ends up in someone else's pocket. Shirking is not an innate tendency toward laziness, but rather the way people are under capitalism. Any complex society will have people with different, sometimes-conflicting interests who need to cooperate toward common goals. Humans have embarked upon and accomplished projects in common, from the mundane to the spectacularly ambitious, long before the advent of capitalism and its subtly coercive labor market—indeed, often involving much more explicit coercion. Across history, however, people have also found ways to plan and act together without bosses to tell them what to do.

In response to any mention of durable human cooperation that is not mediated by markets, in particular by the undisguised incentives provided by the labor market—at their most basic, work or starve—defenders of the market system often bring up the notion of the "tragedy of the commons." The phrase, coined by ecologist Garrett Hardin in a 1968 article in the journal *Science*, refers to a shared resource inevitably depleted through overuse by individuals acting in their self-interest. The prototypical commons employed to illustrate this tragedy is a plot of open, shared pastureland in a village. If farmers only look out for the cows that are theirs, rather than the entire pasture, each will allow their cows to overgraze, and the land shared in common will quickly turn to dust.

Over the course of her long career, Elinor Ostrom, the only woman to win the "not really a Nobel" prize in economics in its fifty-year existence, did much to debunk this crude story. She compiled evidence of groups

stewarding common resources and found that in many cases, the commons not only survived but thrived. Rather than being overrun by unthinking self-interest, shared resources were in reality often governed by complex sets of social rules established over time. Ostrom studied actual shared pasture land in Swiss alpine villages and found it had been preserved for common use for over 500 years. Based on this and other case studies, Ostrom went on to identify conditions that helped protect common resources—among them, participation in decision making by users of the resources, the capacity for monitoring usage, meaningful social sanctions and conflict-resolution mechanisms.

Findings that question the tragedy of the commons, just like the idea of planning itself, can be initially jarring. It is an implicit belief of our age that the only real incentives are pecuniary ones—that despotism is a necessary part of work, and that it is largely out of fear of losing their incomes that people work toward common goals. However, this is not human reality but capitalist reality. While there will always be work that needs doing, there are many ways to organize that work—to plan it and to ensure that it is done. In practice, the commons need not be tragic.

Even within capitalism, studies have shown that a flatter hierarchy makes for better teamwork and greater productivity. Similarly, even just handing all day-to-day operational decision making to the workers doing the work, while leaving only strategic decision making to managers, can boost productivity. Remarkable what giving people more direct decision making over the work processes does! A socialized and truly democratized economy—whether via worker representatives, community councils or more direct forms of democracy—would offer meaningful self-management with no need for illegitimate power of one human over others. In the meantime, simply expanding trade union membership pushes back against the islands of tyranny, giving workers at least some minimal levels of input into working conditions today and laying the foundation for a more thoroughgoing democracy in the future.

Get the Machine before It Gets You

Today, after decades of Hayek-inspired reforms in parliaments and numerous campaigns of outright intimidation in workplaces, union membership is

stagnant or in decline, while democracy in the workplace remains a more distant dream than ever. We are told to celebrate more "flexible" work, to revel in the new freedom to change jobs frequently. Yet despite the transformations wrought by outsourcing and the breakdown of supply chains into smaller pieces, most people remain in stable but crappy jobs in which they have little say over how they work. Despite all the enthusiasm about markets and choice, planning remains the modus operandi of business.

What *has* changed is that the advent of the information technology age has permitted the capture of vast stores of information. What do Facebook and Google do? They prod us, gently and with our own collusion, to reveal information about ourselves. Their business model is the economics of information, come to life. For now, they use the accumulated data to sell ad space—who knew the epitome of high technology would be getting the right people to see ads for novelty "I have a Polish husband and I know how to use him" T-shirts?—but the possibilities are much broader.

Uber and other media darlings of the "sharing economy" combine sussing out information with finding new ways to lower transaction costs. Good capitalists that they are, they're doing it at the expense of workers and democracy (and other capitalists, namely the venture financiers who continue to pump money into a business like Uber even though it has so far failed to turn a profit). Uber's rapid expansion stems in large part from its army of well-paid lobbyists, who in turn cajole and threaten city governments behind closed doors into cutting regulations around taxi monopolies.

Uber's drivers, on the other hand, are poorly-paid "contractors." No longer classified as workers (except in the UK where courts reinstated their rights as workers), they can make below minimum wage and have few labor rights. As with more and more workers in a range of sectors, they are under constant, nigh on panoptical, surveillance via data. Uber uses a five-star driver rating system in which drivers must maintain an average rating of 4.6 stars to keep driving for the company. Uber can "suggest" certain norms for its drivers to follow (how much to smile, what kind of extra services to offer, and so on), but in reality it is the risk of even one bad rating that quickly prods them to fall into line. Yet there is no top-down rule; when businesses can constantly collect and analyze information, strict management happens from the bottom up. Uber's business model is to use the economics of information to do more than just sell ad space. The company's ability to make people do things without telling them explicitly is not unique and is but

a refinement of capitalism's ability to make people complicit in their own unfreedom—a refinement made possible by a greater amount of and greater control over information.

On the other hand, rather than the herald of dystopian workplaces everywhere, Uber is also a natural candidate for a worker co-op. All that Uber provides, after all, is an app; the company is nothing but a middleman. A cooperatively owned network of drivers using a similar app could set pay rates and work rules democratically, in the here and now. A drivers' co-op would be far superior to the venture capital–fueled behemoth we have today, even if this is a form of enterprise that, while introducing more workplace democracy than is normally possible under capitalism in the short run, is still subject to the same profit-seeking imperatives as any firm within capitalism—imperatives that will prompt self-exploitation in order to compete with other enterprises, thus ultimately undermining these very same democratic impulses.

Similarly, social networks could be run as public utilities rather than as private monopolies—remember that we created public electricity or water works after the failures of nineteenth-century robber baron capitalism. One of the big questions of the twenty-first century will be, who owns and controls the data that is quickly becoming a key economic resource? Will it be the fuel for democratic planning, or instead for a new more authoritarian capitalism? These questions require that we recognize the immense challenges posed by data-driven twenty-first-century capitalism: How could we nationalize multinational corporations that span and disregard national borders, and often play jurisdictions off of one another? How would we ensure privacy with so much data under collective, state control?

Privately held data is making possible more efficient production, but at the same time it is enabling closer supervision, and modern corporate planning is only starting to take advantage of all this newly available information. One outcome is illusory freedom for workers. If we constantly produce information both at and outside of work, we don't need to be supervised so directly—but the boss is still watching, and doing so more closely than has ever before been possible. Data and metrics speak for themselves: managers can see how many parts a worker assembled per minute or how many packages a driver delivered per hour.

Increasing self-management at work—ostensibly without managers, but still closely surveilled—is a symptom of bigger changes. As wages, both in

the United States and across much of the global North, have grown at glacial pace or outright stagnated since the 1970s, workers have taken on more personal debt just to keep up. At the same time, governments have cut public benefits, leaving workers more vulnerable when they are laid off or injured. Even Alan Greenspan, the former head of the US Federal Reserve, called today's workers "traumatized." Translated, this means that pressures to fall into line now exist outside the explicit top-down hierarchies.

Capitalism is stuck with planning even though it regularly transmogrifies its techniques of planning. Today, capitalist planning exists both in the old, hierarchical sense that Coase studied as well as in new, more roundabout ways that take cues from the economics of information.

Opening the Gates to the Future

There's an old quip among historians of economics that a PhD-level microeconomics textbook from the 1960s could be mistaken for a textbook at the department of planning at a university in Havana. In the microeconomics textbook, the free market generates the prices that dictate how much of everything is produced and how things are distributed; in the planning textbook, a planner solves the same equations by coming up with the equivalent proportions of production and distribution. Oskar Lange's version of socialism and the economic orthodoxy of the twentieth century shared the same flawed assumptions. Over time, as outlined in this chapter, many poked holes in these assumptions: Markets are costly, said Coase. Human beings are not infinitely powered, all-knowing calculators, argued Stiglitz. Even Hayek was right: capitalism is dynamic, not static, and rarely in the sort of equilibrium imagined by Lange and conventional economics.

But the economics of information also challenges Hayek's counterargument to Lange, that the market is the only means we have to produce all the information that planning would require in the first place. For markets sometimes fail to discover the right information, and that which they do reveal can be false. Also, the enormous amount of economic activity that continues to take place outside the market—within the black boxes we call Walmart or Amazon or General Motors—is evidence against Hayek. At the same time, the rise of information technology shows just how much information it is possible to have at our fingertips. Hayek describes prices as

“a system of telecommunications”; today, we have telecommunications far more precise and powerful that can communicate information directly without it being mediated by prices. Hayek’s arguments may have worked to disarm some of Lange’s vision for planning, but they shouldn’t stop contemporary socialists from arguing for democratic planning that is also a process of discovery.

Economics, if it is to be of use to those who desire an egalitarian society, needs to leave behind fantasy worlds. Paul Samuelson, one of the most influential mainstream economists of the postwar era and author of the economics textbook used in most graduate programs from the 1950s well into the 1970s, observed that in the idealized vision that animated both sides of the calculation debate, it doesn’t matter whether capital hires labor, or labor hires capital. The dense web of abstractions completely obscures what it means to be a boss or a worker, an owner of resources or an owner of a body and mind that can be put to work for a wage.

The economist Duncan Foley describes this lacuna in the calculation debate: “The real import of the historical social choice between socialism and capitalism is precisely what is left out of the socialist calculation debate: the social relations through which people organize themselves to produce.” When we say that we are interested in how things are distributed, we mean that we are interested in how society is organized. Who makes the orders, and who follows them? What counts as “work,” and what is part of the household? Who rears the children, and who does the dishes?

These are only some of the big questions with which any economics of equality will have to grapple. Planning is not only possible, but is already all around us, albeit in hierarchical and undemocratic forms. What a very different, democratic planning will look like is a question a new generation of progressive economists needs to begin today to discuss, debate and test through modeling.

But to the question of whether information should be discovered and created via a system that inevitably creates huge social disparities while depriving a majority of people of a say in how they work, or rather via one of democratic deliberation that fosters equality, the answer should be obvious.